25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Matrices, Determinants & 2D/3D Vectors Practice Exercises

1 MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

A [3, 4]
B [5, 8]
C [-3, -4]
D [4, 3]

2 MCQ

A translation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation is represented by $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$. Find the values of a and b.

A a = -5, b = 0
B a = 5, b = 0
C a = 0, b = -5
D a = 0, b = 5

3 MCQ

A linear transformation in the 2D plane is represented by the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of a, b, c, and d, and then calculate the sum $a + b + c + d$.

A 9
B 11
C 7
D 5

4 MCQ

A matrix $M = \begin{pmatrix} k & 2 \\ 3 & 1 \end{pmatrix}$ transforms a point $P(1, 2)$ to $P'(5, 5)$. Find the value of k.

A 1
B 2
C 3
D 4

5 MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. When the matrix A is applied to the vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$, the resulting vector is $A\mathbf{v} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$. What is the original vector $\mathbf{v}$?

A $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$

B $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

C $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$

D $\begin{pmatrix} 12 \\ 27 \end{pmatrix}$

6 MCQ

Consider a linear transformation T represented by the matrix $A = (2 \quad 1\ 1 \quad 3)$. Let $P = \begin{pmatrix} x \\ y \end{pmatrix}$ be a point in the plane. The transformation maps P to P'. If the area of the region transformed by T from a unit square in the xy-plane is A, and the determinant of A is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is 5.

A 5

B 2

C 1

D 10

7 MCQ

Consider the transformation grid below. If point A has coordinates $(1, 2)$, what are the coordinates of its image, A', after the transformation shown?

A (-4, 2)

B (-2, 1)

C (2, -4)

D (1, -2)

8 MCQ

If $A = \begin{bmatrix} -4 & -3 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 1 \\ -3 & -2 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -8 & -2 \\ -3 & -2 \end{bmatrix}$

B $\begin{bmatrix} 0 & -4 \\ 3 & 2 \end{bmatrix}$

C $\begin{bmatrix} -8 & -1 \\ -3 & -2 \end{bmatrix}$

D $\begin{bmatrix} -8 & -2 \\ -4 & -2 \end{bmatrix}$

9 MCQ

If $A = \begin{bmatrix} -3 & -5 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 0 \\ 3 & 3 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 1 & -5 \\ -3 & -1 \end{bmatrix}$

B $\begin{bmatrix} -7 & -5 \\ 3 & 5 \end{bmatrix}$

C $\begin{bmatrix} -7 & -4 \\ 3 & 5 \end{bmatrix}$

D $\begin{bmatrix} -7 & -5 \\ 2 & 5 \end{bmatrix}$

10 MCQ

If $A = \begin{bmatrix} 1 & -1 \\ -4 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ -2 & -2 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 4 & -2 \\ -6 & -7 \end{bmatrix}$

B $\begin{bmatrix} -2 & 0 \\ -2 & -3 \end{bmatrix}$

C $\begin{bmatrix} 4 & -1 \\ -6 & -7 \end{bmatrix}$

D $\begin{bmatrix} 4 & -2 \\ -7 & -7 \end{bmatrix}$

11 MCQ

If $A = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -5 \\ -3 & -1 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -8 & 8 \\ 5 & 0 \end{bmatrix}$

B $\begin{bmatrix} -2 & -2 \\ -1 & -2 \end{bmatrix}$

C $\begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}$

D $\begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}$

12 MCQ

If $A = \begin{bmatrix} -5 & 5 \\ 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -5 & -5 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -4 & 9 \\ -6 & 0 \end{bmatrix}$

B $\begin{bmatrix} -4 & 9 \\ -5 & 0 \end{bmatrix}$

C $\begin{bmatrix} -6 & 1 \\ 5 & 10 \end{bmatrix}$

D $\begin{bmatrix} -4 & 10 \\ -5 & 0 \end{bmatrix}$

13 MCQ

If $A = \begin{bmatrix} -2 & 0 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -3 & 1 \\ 2 & 4 \end{bmatrix}$

B $\begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}$

C $\begin{bmatrix} -3 & 2 \\ 2 & 4 \end{bmatrix}$

D $\begin{bmatrix} -3 & 1 \\ 1 & 4 \end{bmatrix}$

14 MCQ

If $A = \begin{bmatrix} 2 & -5 \\ -3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & -4 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 2 & -6 \\ -4 & 0 \end{bmatrix}$

B $\begin{bmatrix} 2 & -3 \\ -2 & -8 \end{bmatrix}$

C $\begin{bmatrix} 2 & -4 \\ -3 & -8 \end{bmatrix}$

D $\begin{bmatrix} 2 & -4 \\ -2 & -8 \end{bmatrix}$

15 MCQ

If $A = \begin{bmatrix} 5 & 4 \\ 0 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 2 & 1 \\ -1 & -5 \end{bmatrix}$

B $\begin{bmatrix} 8 & 8 \\ 1 & -3 \end{bmatrix}$

C $\begin{bmatrix} 8 & 7 \\ 1 & -3 \end{bmatrix}$

D $\begin{bmatrix} 8 & 7 \\ 0 & -3 \end{bmatrix}$

16 MCQ

If $A = \begin{bmatrix} 3 & -5 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ -2 & -2 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 3 & -5 \\ -2 & -1 \end{bmatrix}$

B $\begin{bmatrix} 3 & -6 \\ -3 & -1 \end{bmatrix}$

C $\begin{bmatrix} 3 & -6 \\ -2 & -1 \end{bmatrix}$

D $\begin{bmatrix} 3 & -4 \\ 2 & 3 \end{bmatrix}$

17 MCQ

If $A = \begin{bmatrix} 3 & -5 \\ 1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -1 \\ -5 & 5 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 8 & -6 \\ -4 & 0 \end{bmatrix}$

B $\begin{bmatrix} -2 & -4 \\ 6 & -10 \end{bmatrix}$

C $\begin{bmatrix} 8 & -5 \\ -4 & 0 \end{bmatrix}$

D $\begin{bmatrix} 8 & -6 \\ -5 & 0 \end{bmatrix}$

18 MCQ

If $A = \begin{bmatrix} -3 & 4 \\ 5 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 2 & 4 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -5 & 4 \\ 3 & -7 \end{bmatrix}$

B $\begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$

C $\begin{bmatrix} -1 & 4 \\ 6 & 1 \end{bmatrix}$

D $\begin{bmatrix} -1 & 4 \\ 7 & 1 \end{bmatrix}$

19 MCQ

If $A = \begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ -5 & 3 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix}$

B $\begin{bmatrix} 8 & -2 \\ -7 & 4 \end{bmatrix}$

C $\begin{bmatrix} 8 & -1 \\ -7 & 4 \end{bmatrix}$

D $\begin{bmatrix} 8 & -2 \\ -8 & 4 \end{bmatrix}$

20 MCQ

If $A = \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 2 & -3 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 4 & 2 \\ 1 & 2 \end{bmatrix}$

B $\begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix}$

C $\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$

D $\begin{bmatrix} -2 & 1 \\ -3 & 8 \end{bmatrix}$

21 MCQ

If $A = \begin{bmatrix} -6 & -2 \\ 3 & -3 \end{bmatrix}$, find $-4A$.

A $\begin{bmatrix} 20 & 8 \\ -12 & 12 \end{bmatrix}$

B $\begin{bmatrix} -24 & -8 \\ 12 & -12 \end{bmatrix}$

C $\begin{bmatrix} 20 & 4 \\ -16 & 8 \end{bmatrix}$

D $\begin{bmatrix} 24 & 8 \\ -12 & 12 \end{bmatrix}$

22 MCQ

Suppose that $A = \begin{bmatrix} -4 & -6 \\ 1 & -2 \end{bmatrix}$, find $-2A$.

A $\begin{bmatrix} 8 & 12 \\ -2 & 4 \end{bmatrix}$

B $\begin{bmatrix} -8 & -12 \\ 2 & -4 \end{bmatrix}$

C $\begin{bmatrix} 6 & 10 \\ -4 & 2 \end{bmatrix}$

D $\begin{bmatrix} 6 & 12 \\ -2 & 4 \end{bmatrix}$

23 MCQ

If $A = \begin{bmatrix} -6 & -4 \\ -5 & 6 \end{bmatrix}$, find $-3A$.

A $\begin{bmatrix} 15 & 9 \\ 12 & -21 \end{bmatrix}$

B $\begin{bmatrix} 15 & 12 \\ 15 & -18 \end{bmatrix}$

C $\begin{bmatrix} 18 & 12 \\ 15 & -18 \end{bmatrix}$

D $\begin{bmatrix} -18 & -12 \\ -15 & 18 \end{bmatrix}$

24 MCQ

Assuming that $A = \begin{bmatrix} -3 & -3 \\ 6 & -1 \end{bmatrix}$, find $-2A$.

A $\begin{bmatrix} 4 & 4 \\ -14 & 0 \end{bmatrix}$

B $\begin{bmatrix} 6 & 6 \\ -12 & 2 \end{bmatrix}$

C $\begin{bmatrix} 4 & 6 \\ -12 & 2 \end{bmatrix}$

D $\begin{bmatrix} -6 & -6 \\ 12 & -2 \end{bmatrix}$

25 MCQ

If $A = \begin{bmatrix} -1 & 2 \\ 4 & 4 \end{bmatrix}$, find $-4A$.

A $\begin{bmatrix} -4 & 8 \\ 16 & 16 \end{bmatrix}$

B $\begin{bmatrix} 4 & -8 \\ -16 & -16 \end{bmatrix}$

C $\begin{bmatrix} 0 & -12 \\ -20 & -20 \end{bmatrix}$

D $\begin{bmatrix} 0 & -8 \\ -16 & -16 \end{bmatrix}$